Nicholas P. Dana, PhD

Entrepreneur, Collaborator, AI & Cross-Industry Innovator

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SUMMARY (Updated October 2026)

Innovative data scientist and AI engineer integrating advanced analytics and AI to accelerate scientific discovery and product innovation. Proven leader in building scalable systems, automating complex pipelines, and translating R&D breakthroughs into operational tools that drive value across business applications and technology sectors.

EXPERIENCE

Off Label Development & Phronesix — Founder & CTO

May 2026 – Present | Remote / Denver, CO

- Founded Off Label Development, an AI consultancy bringing practical AI to small businesses — strategy sessions, workflow automation (n8n, Zapier, custom code), team training, and custom builds (custom GPTs, agents, RAG systems)
- Founded Phronesix, a quantitative consulting practice for biotech, pharma, and healthcare: Bayesian/hierarchical modeling, survival & meta-analysis, causal inference, reproducible data pipelines, study design review, and technical due diligence
- Own strategy, technical architecture, and delivery; every engagement built on explicit assumptions and decision-focused outputs

Elephas Biosciences — Principal AI Scientist & Developer

June 2021 – October 2026 | Remote / Madison, WI

- Architected and deployed ***multi-modal analytics pipelines*** in cloud-integrated ML workflows leveraging image, cytokine, and assay data to predict drug response in *ex vivo* tumor models
- ***Leadership Impact***: Technical lead for SME group (8) developing custom analytics achieving ***>96% training data accuracy*** and ***87% concordance*** with validation samples
- ***Strategic Initiative***: Prototyped analytic methods leading to company's first Lab Derived Test (LDT, link to test offering)
- ***Strategic Initiative***: Led group efforts to develop and implement a testing framework (adapted TOST approach) to validate LDT method-transfer to multiple international sites
- Co-authored multiple conference & BioRxiv publications and abstracts advancing translational oncology (see below)

GeoVisual Analytics — Computer Vision & Machine Learning Scientist

2018 – 2021 | Boulder, CO

- Built and trained multiple ***deep-learning CV models*** for agricultural analytics using drone-based imagery
- Developed ***best-in-class models*** for crop maturation and yield forecasting, leveraging multi-modal data from custom image systems, weather sensors and feeds, and geolocation data

- ***Quantifiable Result***: Achieved ***0-day median error*** in harvest date prediction and ***<5% error*** in crop yield estimation across multiple production sites throughout central California and Arizona

Essen BioScience (Sartorius) — Algorithm Engineer

2016 – 2018 | Ann Arbor, MI

- Designed and implemented segmentation and denoising algorithms enabling automated live-cell image analysis with minimal user input
- ***Quantifiable Result***: Developed a computer vision library for time-series imaging, achieving ***99.5% data fidelity*** at ***14× probabilistic compression efficiency***

CERTIFICATIONS & CREDENTIALS

Post Graduate Program in AI Agents for Business Applications

University of Texas at Austin - McCombs School of Business, May 2026

Accelerated course giving students the technical and conceptual depth to design, deploy, monitor, improve and govern Agentic AI solutions for a variety of business applications

EDUCATION

Doctor of Philosophy, Biomedical Engineering

The University of Texas at Austin, 2016

NIH T32 Bioinformatics Fellowship

Dissertation: *Photoacoustic Imaging for Tissue Characterization and Image Guidance in Cardiovascular Applications*

Bachelor of Science, Physics

University of Utah, 2010

Focus in Medical Physics, Mathematics and Chemistry

SELECTED PUBLICATIONS

- Dana N. *et al.*, ***"Prediction of immunotherapy response using live tumor fragments from routine clinical biopsies"*** Manuscript in review (link)
- Ramasubramanian T.S. *et al.*, ***"A live tumor fragment platform to assess immunotherapy response in core needle biopsies,"***, *Journal of Translational Medicine* 2025 (link)
- Dana N. *et al.*, ***"Development and characterization of cytokine-based classifiers to predict response to immunotherapy using an ex vivo live tumor fragment platform"***, *Society of Immunotherapy in Cancer 2025 Poster* (link)
- Dana N. *et al.*, ***"Optimization of dual-wavelength intravascular photoacoustic imaging of atherosclerotic plaques using Monte Carlo optical modeling"***, *Journal of Biomedical Optics*, 2017 (link)
- Dana N. *et al.*, ***"Overview of Photoacoustic Imaging"***, *Advances in Medical Physics*, 2016 edition.

TECHNICAL SKILLS

Tools: Python, SQL, C++, C#, R, MATLAB, Docker

AI / Data Science: GenAI, AgenticAI, LangChain / LangGraph, PyTorch / Tensorflow, Deep Learning, Scikit, Pandas, Bayesian Methods, Dimensionality Reduction, Unsupervised Learning, Computer Vision

Domain Expertise: Optical Imaging, Assay Development and Validation, Multi-modal Analysis, High-dimensional Analysis, Bioinformatics, Computational Modeling